

SQL PROJECT(PIZZA TABLE)

INTRODUCTION

The Pizza Sales dataset contains transactional data related to pizza orders from a pizza store. It includes information about orders, pizza types, quantities sold, prices, order dates, and order times. This dataset helps in analyzing sales performance, customer ordering patterns, and product popularity.

OBJECTIVES OF PIZZA SALES PROJECT

- **TO ANALYZE PIZZA SALES DATA** AND UNDERSTAND OVERALL BUSINESS PERFORMANCE.
- **TO IDENTIFY THE MOST POPULAR PIZZA TYPES AND CATEGORIES** BASED ON CUSTOMER ORDERS.
- **TO CALCULATE TOTAL REVENUE AND SALES TRENDS** USING SQL QUERIES.
- **TO UNDERSTAND CUSTOMER ORDERING PATTERNS** BY DAY AND TIME.
- **TO FIND TOP-SELLING PIZZAS AND HIGH-VALUE ORDERS** TO SUPPORT BUSINESS INSIGHTS.
- **TO PRACTICE DATA ANALYSIS SKILLS** USING SQL AND VISUALIZE INSIGHTS FOR BETTER DECISION-MAKING.

PIZZA SALES

- **RETRIEVE ALL ORDERS PLACED ON 2015-01-01.**
- **LIST ALL DISTINCT PIZZA CATEGORIES AVAILABLE.**
- **FIND TOTAL NUMBER OF PIZZAS ORDERED.**
- **GET THE TOP 5 MOST FREQUENTLY ORDERED PIZZA TYPES.**
- **SHOW ORDER_ID AND TOTAL PIZZAS ORDERED IN EACH ORDER.**

PIZZA SALES

- **FIND THE NUMBER OF ORDERS PLACED IN EACH HOUR OF THE DAY.**
- **FIND THE DAY OF THE WEEK ON WHICH THE MAXIMUM NUMBER OF ORDERS WERE PLACED.**
- **FIND THE EARLIEST AND LATEST ORDER TIME FOR EACH DAY.**
- **FIND THE TOTAL REVENUE GENERATED FROM EACH PIZZA TYPE.**
- **WHICH PIZZA CATEGORY GENERATED THE MAXIMUM REVENUE?**
- **FIND THE AVERAGE NUMBER OF PIZZAS ORDERED PER ORDER.**
- **LIST THE TOP 3 ORDERS WHICH SPENT THE MOST.**
- **FIND THE CUMULATIVE REVENUE GENERATED THROUGHOUT THE DAY (RUNNING TOTAL).**
- **FIND THE PERCENTAGE CONTRIBUTION OF EACH PIZZA CATEGORY TO TOTAL REVENUE.**

1-Retrieve all orders placed on QUERY -

```
select * from ORDERS where date = '2015-01-01'
```

order_id	date	time
1	2015-01-01	11:38:36.0000000
2	2015-01-01	11:57:40.0000000
3	2015-01-01	12:12:28.0000000
4	2015-01-01	12:16:31.0000000
5	2015-01-01	12:21:30.0000000
6	2015-01-01	12:29:36.0000000
7	2015-01-01	12:50:37.0000000
8	2015-01-01	12:51:37.0000000
9	2015-01-01	12:52:01.0000000
10	2015-01-01	13:00:15.0000000
11	2015-01-01	13:02:59.0000000
12	2015-01-01	13:04:41.0000000

2-List all **distinct** pizza categories available.

QUERY -

```
select distinct(category) as pizzas_category  
from pizzatypes
```

	pizzas_category
1	Chicken
2	Classic
3	Supreme
4	Veggie

3-Find total number of pizzas ordered.

QUERY -

```
select sum(quantity) as order_count from  
[ORDERS _DETAILS]
```

	order_count
1	49574

4-Get the top 5 most frequently ordered pizza types.

QUERY -

```
SELECT TOP 5 pt.name, SUM(od.quantity)
AS total_orders
FROM [ORDERS _DETAILS] od
JOIN pizzas p
ON od.pizza_id = p.pizza_id
JOIN pizzatypes pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY total_orders DESC
```

	name	total_orders
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

5-Show order_id and total pizzas ordered in each order.

QUERY -

```
select o.order_id, COUNT(od.quantity) as  
order_count from ORDERS o join [ORDERS  
_DETAILS] as od on  
o.order_id = od.order_id  
group by o.order_id
```

	order_id	order_count
1	2917	2
2	5834	2
3	11407	3
4	14324	4
5	19897	4
6	1566	1
7	7185	1
8	10056	1
9	15675	2
10	18546	1
11	46	3
12	215	1

6-Find the number of orders placed in each hour of the day.

QUERY -

```
select date, DATEPART(hour,time) as  
hours,COUNT(order_id) as total_order from  
ORDERS  
group by date, DATEPART(hour,time)  
order by date, hours
```

	date	hours	total_order
1	2015-01-01	11	2
2	2015-01-01	12	7
3	2015-01-01	13	10
4	2015-01-01	14	7
5	2015-01-01	15	7
6	2015-01-01	16	4
7	2015-01-01	17	8
8	2015-01-01	18	8
9	2015-01-01	19	6
10	2015-01-01	20	5
11	2015-01-01	21	2
12	2015-01-01	22	3

7-Find the **day of the week** on which the maximum number **of** orders were placed.

QUERY -

```
select datename(WEEKDAY,date)as weekday,  
COUNT(order_id) as order_count from ORDERS  
group by datename(WEEKDAY,date)  
order by order_count desc
```

	weekday	order_count
1	Friday	3538
2	Thursday	3239
3	Saturday	3158
4	Wednesday	3024
5	Tuesday	2973
6	Monday	2794
7	Sunday	2624

8-Find the earliest and latest order time for each day.

QUERY -

```
select date,  
MIN(time) as earliest_time,  
MAX(time) as latest_time  
from ORDERS  
group by date
```

	date	earliest_time	latest_time
1	2015-02-14	11:48:35.0000000	22:52:37.0000000
2	2015-01-22	11:34:52.0000000	23:02:31.0000000
3	2015-12-20	11:55:46.0000000	22:25:20.0000000
4	2015-04-01	11:17:06.0000000	22:41:48.0000000
5	2015-03-09	11:34:01.0000000	21:58:39.0000000
6	2015-07-27	11:34:12.0000000	22:23:58.0000000
7	2015-08-02	12:01:58.0000000	22:12:31.0000000
8	2015-08-19	11:24:03.0000000	22:30:27.0000000
9	2015-08-25	11:46:31.0000000	22:17:58.0000000
10	2015-06-11	11:25:26.0000000	21:43:38.0000000
11	2015-11-27	11:21:54.0000000	22:36:01.0000000
12	2015-07-04	12:20:55.0000000	23:04:40.0000000

9-Find the total revenue generated from each pizza type.

QUERY -

```
select pt.name, SUM(od.quantity*p.price) as  
revenue from pizzatypes pt join PIZZAS p on  
pt.pizza_type_id = p.pizza_type_id join  
[ORDERS _DETAILS] od on  
p.pizza_id = od.pizza_id  
group by pt.name
```

	name	revenue
1	The Napolitana Pizza	24087
2	The Barbecue Chicken Pizza	42768
3	The Pepper Salami Pizza	25529
4	The Five Cheese Pizza	26066.5
5	The Mexicana Pizza	26780.75
6	The Brie Carre Pizza	11588.4998130798
7	The Classic Deluxe Pizza	38180.5
8	The Italian Vegetables Pizza	16019.25
9	The Pepperoni Pizza	30161.75
10	The Thai Chicken Pizza	43434.25
11	The Sicilian Pizza	30940.5
12	The Soppressata Pizza	16425.75

10--Which pizza category generated the maximum revenue?

```
QUERY -
select top 1 pt.category,
SUM(od.quantity * p.price) as revenue
from pizzatypes pt
join pizzas p
    on pt.pizza_type_id = p.pizza_type_id
join [ORDERS _DETAILS] od
    on p.pizza_id = od.pizza_id
group by pt.category
```

	category	revenue
1	Chicken	195919.5

11-Find the average number of pizzas ordered per order.

QUERY -

```
with cte as(  
select order_id, SUM(quantity) as  
total_quantity from [ORDERS _DETAILS]  
group by order_id)  
select AVG(total_quantity) as avg_no from  
cte
```

	avg_no
1	2

12-List the top 3 orders which spent the most.

QUERY -

```
select top 3  
od.order_id, SUM(od.quantity*p.price) as  
total_qty from [ORDERS _DETAILS] od join  
PIZZAS p on  
od.pizza_id = p.pizza_id  
group by od.order_id  
order by total_qty desc
```

	order_id	total_qty
1	18845	444.200000762939
2	10760	417.14999961853
3	1096	285.150001525879

13-Find the cumulative revenue generated throughout the day (running total).

QUERY -

```
select date,o.time,sum(od.quantity*p.price)
as total_revenue,
SUM(sum(od.quantity*p.price)) over(partition
by o.date order by o.time) as cumsum
from [ORDERS _DETAILS] od join ORDERS o on
od.order_id = o.order_id join PIZZAS p on
od.pizza_id = p.pizza_id
group by date,o.time
```

	date	time	total_revenue	cumsum
1	2015-01-01	11:38:36.0000000	13.25	13.25
2	2015-01-01	11:57:40.0000000	92	105.25
3	2015-01-01	12:12:28.0000000	37.25	142.5
4	2015-01-01	12:16:31.0000000	16.5	159
5	2015-01-01	12:21:30.0000000	16.5	175.5
6	2015-01-01	12:29:36.0000000	24.75	200.25
7	2015-01-01	12:50:37.0000000	12.5	212.75
8	2015-01-01	12:51:37.0000000	12.5	225.25
9	2015-01-01	12:52:01.0000000	143.25	368.5
10	2015-01-01	13:00:15.0000000	41	409.5
11	2015-01-01	13:02:59.0000000	73.5	483
12	2015-01-01	13:04:41.0000000	70.75	553.75

15--Find the percentage contribution of each pizza category to total revenue.

```
QUERY -  
SELECT  
    pt.category,  
    SUM(od.quantity * p.price) AS category_revenue,  
    (SUM(od.quantity * p.price) * 100.0) /  
    SUM(SUM(od.quantity * p.price)) OVER() AS  
percentage_contribution  
FROM [ORDERS _DETAILS] od  
JOIN pizzas p  
    ON od.pizza_id = p.pizza_id  
JOIN pizzatypes pt  
    ON p.pizza_type_id = pt.pizza_type_id  
GROUP BY pt.category  
ORDER BY percentage_contribution DESC
```

	category	category_revenue	percentage_contribution
1	Classic	220053.100021362	26.9059602306976
2	Supreme	208196.99981308	25.4563112111462
3	Chicken	195919.5	23.9551375322885
4	Veggie	193690.451004028	23.6825910258677

CONCLUSION

The Pizza Sales Analysis project helped in understanding customer ordering behavior and overall business performance using SQL queries. By analyzing the dataset, we identified the most popular pizza types, highest revenue-generating categories, and peak order times. The analysis also highlighted top orders with the highest spending and the contribution of each pizza category to total revenue.